



ALFRED UNIVERSITY

SCHOOL OF ENGINEERING AND APPLIED TECHNOLOGY

Department of Applied Electronics and Embedded Technology

Embedded Systems + CETa Career Transition Program

Course Syllabus & Student Handbook

AU-ESET 301 | Section 001 | Fall 2026 - Spring 2027

STUDENT	William A. Green IV
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Personal independent-study simulation. Alfred University is a fictional institution and this document is not an accredited university record.

CURRENT WEBSITE USE - v15.8

Use Academic System Guide Revision 2.1 for today's workflow. Study owns daily work; Week owns weekly instruction; Calendar owns scheduling; Progress owns completion and backups. Standards and Mastery summarize current evidence. Earlier gradebook, hours-log and manual mastery instructions do not apply.



Program status

This is a structured, independent, non-credit professional study program designed to feel and operate like a real course. It is not an accredited college course and does not award academic credit. Its purpose is to create external structure, deadlines, grading, labs, mastery gates and career-ready evidence.

Course at a Glance

Course code	AU-ESET 301 (personal non-credit sequence)
Course title	Embedded Systems + CETa Career Transition
Course dates	September 8, 2026 - April 10, 2027
Delivery	Independent study + scheduled lab blocks + AI tutoring/office-hours support
Primary objective	Reach bridge-job application readiness with demonstrable electronics, embedded, test and R&D-lab evidence
Certification objective	Reach CETa readiness using current ETA scope and two current practice runs at $\geq 85\%$
Capstone	STM32-based Project 1 with sensor/interface, structured firmware, Python test/logger and documented troubleshooting
Application gate	Week 30 green-light; Week 31 begins the application sprint

Academic Calendar

Term	Weeks	Dates	Primary emphasis
Term I - Foundations	Weeks 1-15	Sep 8 - Dec 19, 2026	Electronics, instruments, circuits, semiconductors, digital logic, schematics, C and embedded C
Winter Embedded Intensive	Weeks 16-18	Dec 21, 2026 - Jan 9, 2027	STM32, SWD, UART, Python serial, I2C/SPI, logic analyzer, sensors/actuators
Term II - Applied Transition	Weeks 19-31	Jan 12 - Apr 10, 2027	CETa completion, Project 1, hardware test automation, root-cause analysis, role proof, portfolio and applications



1. Course Description

This course sequence turns the bridge-job readiness plan into a school-like academic program. The course begins at zero assumptions, builds the electronics foundation required for hands-on work, develops C and STM32 competence, integrates professional test equipment and communication protocols, prepares for CETa, and culminates in an employer-facing embedded hardware project and bridge-job application campaign.

The course is intentionally mastery-based. Calendar dates create deadlines, but a topic is not considered complete merely because the assigned reading or video was finished. Completion requires evidence: calculations, code, hardware, measurements, debugging, documentation or a passing mastery check as appropriate.

Core learning cycle

LEARN -> RETRIEVE -> CALCULATE/CODE -> BUILD/MEASURE -> DEBUG -> DOCUMENT. Watching or reading alone is exposure, not mastery.

2. Learning Outcomes

- Explain and calculate foundational DC and AC electronics concepts used in electronics technician and embedded-hardware work.
- Use a DMM, bench power supply and oscilloscope safely and productively on current-limited low-voltage circuits.
- Read schematics, wiring/block diagrams and datasheets well enough to trace power, signals, interfaces and likely faults.
- Identify and reason about passive components, diodes, transistors, MOSFETs, power supplies, op-amps and digital logic at bridge-job depth.
- Write explainable modular C code using functions, arrays, structs, enums, pointers, bitwise operations and embedded-C practices.
- Bring up and debug an STM32-class microcontroller using GPIO, timers/interrupts, UART and SWD debugging.
- Implement and troubleshoot at least one board-level serial bus, I2C or SPI, and interpret logic-analyzer captures.
- Use Python to communicate with hardware, log data and build repeatable PASS/FAIL test automation.
- Apply systematic troubleshooting: reproduce, measure, isolate, identify root cause, fix and verify.
- Demonstrate soldering/rework, wiring/harness and lab-workmanship evidence appropriate to R&D/electronics technician roles.
- Produce professional documentation: requirements, architecture, lab notes, test reports, failure reports and portfolio evidence.
- Present a credible bridge-job application package and explain technical work in interviews without exaggerating skill level.



3. Prerequisites

No prior electronics mastery is assumed. General computer literacy and willingness to use the command line are sufficient. Prior Python experience is helpful but not required. All personal bench work should remain low-voltage and current-limited unless supervised by a qualified environment with appropriate procedures.

4. Required Course Materials

- **Associate CET Study Guide, 6th Edition:** Primary CETa textbook; use printed page numbers. Current ETA competencies and errata control conflicts.
- **ETA CETa Competencies:** Official exam scope [Open resource](#)
- **ETA Common Formulas:** Official formula reference [Open resource](#)
- **ETA Electronics / CETa Hub:** Current certification information [Open resource](#)
- **All About Circuits Textbook:** Primary free written electronics reference [Open resource](#)
- **Beej's Guide to C:** Primary C fundamentals reference [Open resource](#)
- **Modern Embedded Systems Programming:** Embedded C / ARM companion course [Open resource](#)
- **STM32CubeIDE Basics MOOC:** Official STM32 setup/GPIO/debug training [Open resource](#)
- **STM32CubeMX & CubeHAL Basics MOOC:** Official STM32 peripheral/HAL training [Open resource](#)
- **SparkFun Serial / I2C / SPI Tutorials:** Protocol learning and practical reference [Open resource](#)
- **Saleae Support:** Logic analyzer and protocol decoding reference [Open resource](#)
- **pySerial:** Python serial automation documentation [Open resource](#)
- **pytest:** Automated test framework documentation [Open resource](#)
- **Matplotlib Quick Start:** Test-data visualization [Open resource](#)
- **Tektronix XYZs of Oscilloscopes:** Oscilloscope training [Open resource](#)
- **Keysight Bench Power Supply Basics:** Bench supply training [Open resource](#)
- **NASA-STD-8739.4:** Harness/wiring workmanship reference [Open resource](#)

5. Weekly Course Format

The default academic week uses four kinds of work:

- **Learn / Map:** New concepts, assigned book sections and primary resources.
- **Retrieve / Calculate:** Closed-note recall, formulas, fresh problems and CETa-aligned checks.
- **Apply / Simulate / Code:** Use the concept in LTspice, C, Python, STM32 or structured troubleshooting.
- **Saturday Lab / Mastery:** Build, measure, break/fix, document and close the week with an evidence-based mastery decision.



6. Time Commitment & Attendance

Treat scheduled study blocks like class meetings. The course is designed around approximately 7-8 hours per week during the early foundation term and a lighter load during the spring semester, with the exact blocks already carried in the V6.2 calendar.

Attendance standard: complete at least 80% of scheduled study blocks each month. A missed block is not a failure; reschedule it within seven days when practical. Do not create unhealthy make-up marathons. College coursework and unavoidable work/family obligations take priority over preserving an arbitrary streak.

7. Grading System

Category	Weight
Knowledge & retrieval checks	15%
Calculations / CETa quizzes	15%
Hands-on labs & practical demonstrations	25%
CETa readiness assessments	10%
Project 1 capstone	20%
Career proof / portfolio / interview preparation	10%
Professional habits & attendance	5%

Grade scale: A = 90-100; B = 80-89.9; C = 70-79.9; D = 60-69.9; F = below 60. Course completion requires a C or better AND all safety-critical/mastery gates below must be satisfied. A high numeric average cannot override an unsafe or missing practical gate.

8. Mastery Gates

When	Gate	Passing evidence
Week 5	Foundation Gate	$\geq 80\%$ mixed mastery + independently calculate/build/measure/fault-isolate a basic circuit.
Week 12	Fabrication + Cumulative Gate	Solder/rework/ESD evidence + cumulative repair.
Week 18	Embedded Interface Core Gate	STM32 + SWD + UART + I2C or SPI + logic analyzer are demonstrable.
Weeks 21-23	CETa Readiness Gate	Repair weak areas and complete two current practice runs at $\geq 85\%$ before calling CETa readiness green.
Week 30	Bridge-Job Application Gate	Common core + Project 1 + one role proof + professional package are demonstrable; begin applications.
Week 31	Application Sprint	Apply, interview, log feedback and convert real market feedback into dated repairs.



9. Major Assignments & Practicals

A01 - Engineering Lab Notebook

Due: Week 1

Initialize Git/GitHub evidence repo and keep dated notes, calculations, measurements, photos, code and troubleshooting records.

P01 - DC Measurement & Fault Practical

Due: Week 5

Calculate, build, measure and fault-isolate a low-voltage circuit; meet the 80% foundation gate.

P02 - Soldering / Rework Practical

Due: Week 12

Demonstrate safe soldering, desoldering, rework and ESD habits; submit before/after evidence.

P03 - Embedded Interface Practical

Due: Week 18

STM32 + UART + I2C or SPI + logic-analyzer proof; demonstrate debugging and interface understanding.

E01 - CETa Readiness Assessment

Due: Weeks 21-23

Repair weak competencies and achieve two current practice runs at 85% or higher before calling CETa readiness green.

D01 - Project 1 Design Review

Due: Week 22

Requirements, architecture, block diagram, BOM, pin map, interfaces, test plan and scope freeze.

C01 - Project 1 Capstone

Due: Weeks 24-28

Build the complete STM32 system, integrate sensor/peripheral, structured firmware, Python logger/test harness and fault handling.

R01 - Root-Cause & Test Report

Due: Week 28

Document at least three failures using reproduce-measure-isolate-root cause-fix-verify and produce a professional test report.



R02 - R&D / Lab Role-Proof Dossier

Due: Week 29

Show instrumentation, solder/rework, harness/termination, bring-up and troubleshooting evidence.

F01 - Career Transition Final

Due: Week 30

Targeted resumes, portfolio/GitHub, 90-second transition story, five technical/behavioral stories and final green-light scorecard.

A02 - Application Sprint

Due: Week 31

Submit targeted applications, track outcomes and convert interview feedback into dated skill repairs.

10. Project 1 Capstone Requirements

- STM32/ARM-class MCU
- At least one real sensor/peripheral using I2C OR SPI
- Digital input with debounce
- Controlled output/actuator or representative load
- UART diagnostic/command interface
- Python host/logger and automated test element
- Timer/interrupt-driven behavior
- Non-blocking state-machine structure
- Fault/timeout handling
- DMM, oscilloscope and logic-analyzer evidence
- Git repository + README + architecture/schematic + test plan + results + photos/demo
- At least three documented troubleshooting stories

11. CETa Readiness Policy

The course uses the Associate CET Study Guide, 6th edition as the primary textbook, but current ETA competencies and current ETA errata control when the older book differs or omits a current topic. The course grade is based on demonstrated learning and readiness; the official certification exam is external to this course. The internal readiness standard is intentionally stricter: two current practice runs at 85% or higher with weak competencies repaired.

12. Academic & Lab Policies

Late work: Weekly work may be completed up to seven days late without penalty if the delay is caused by work, school or family obligations. Major milestone slippage must be explicitly rescheduled, not silently ignored.



Mastery before completion: Do not mark a skill complete because you watched a video, copied code or followed a tutorial. Demonstrate it independently in a fresh example or real lab.

Academic integrity / AI use: AI may be used as a tutor, explainer, debugger and reviewer. It may not substitute for demonstrated understanding. Portfolio artifacts must reflect work you can explain and defend in an interview.

Safety: Personal bench work stays low-voltage/current-limited. Follow manufacturer and official safety procedures for DMM current measurements, bench supplies, oscilloscope grounding, soldering and ESD.

Documentation: Every meaningful lab should leave evidence: notes, calculations, photos, code, measurements, captures or a test report.

Remediation: Yellow = continue with no more than one or two clearly dated repairs. Red = repair a core/safety/practical gap before calling that gate complete.

School priority: When college exams/projects collide with this independent course, college takes priority. Reduce the self-study load rather than sacrificing sleep or academic performance.

13. Instructor Support / Office Hours

Use ChatGPT as your on-demand study coach and office-hours support. Bring the exact week, assignment code, lab result, failed calculation, code error, waveform or job posting. Ask for explanation, a practice check, troubleshooting help or a grading review against the rubric. For objective technical claims, continue using the primary course resources and official/manufacturer documentation as the controlling references.



14. 31-Week Course Schedule

The weekly due point is the end of the scheduled Saturday mastery/lab block unless a major assessment explicitly says otherwise. The calendar remains the operational source for exact session times.

Term I - Foundations

Week / Dates	Topic	Book	Due / Gate	Status
W01 Sep 08 - Sep 12	Electricity Basics + Safety + Ohm/Watt	Chapter 1 pp. 1-6; Chapter 2 pp. 7-15; Chapter 3 pp. 16-24; Chapter 4 pp. 25-34	Weekly mastery	[]
W02 Sep 14 - Sep 19	DC Networks + KVL/KCL + Voltage Dividers + DMM	Chapter 4 pp. 25-34	Weekly mastery	[]
W03 Sep 21 - Sep 26	Test Equipment First: DMM + Bench Supply + Oscilloscope	Chapter 19 pp. 165-175	Weekly mastery	[]
W04 Sep 28 - Oct 03	AC Waveforms + R/C/L Behavior + Frequency/Period/RMS	Chapter 3 pp. 16-24; Chapter 5 pp. 36-44; Chapter 6 pp. 45-52	Weekly mastery	[]
W05 Oct 05 - Oct 10	BUFFER 1: DC/AC Fundamentals Mastery Check	Targeted review / no major new chapter	Foundation Gate	[]
W06 Oct 12 - Oct 17	Reactance + Impedance + Resonance + Filters + Transformers	Chapter 3 pp. 16-24; Chapter 5 pp. 36-44; Chapter 6 pp. 45-52	Weekly mastery	[]
W07 Oct 19 - Oct 24	Passive Components + Magnetism + Motors/Generators + Relays	Chapter 7 pp. 53-58; Chapter 8 pp. 62-69	Weekly mastery	[]
W08 Oct 26 - Oct 31	Semiconductors: Diodes + BJT/FET/MOSFET + SCR/TRIAC	Chapter 9 pp. 70-78; Chapter 10 pp. 79-83	Weekly mastery	[]
W09 Nov 02 - Nov 07	Power Supplies + Rectifiers + Regulation + Protection	Chapter 11 pp. 85-90	Weekly mastery	[]
W10 Nov 09 - Nov 14	Amplifiers + Op-Amps + Gain + Loading	Chapter 12 pp. 91-101	Weekly mastery	[]
W11 Nov 16 - Nov 21	Digital Logic + Number Systems + Boolean + Logic Families	Chapter 16 pp. 132-148	Weekly mastery	[]
W12 Nov 23 - Nov 28	LIGHT / BUFFER 2: Soldering + Git + Cumulative Repair	Chapter 20 pp. 176-180	Fabrication Gate	[]
W13 Nov 30 - Dec 05	Schematics + Block/Wiring Diagrams + Systematic Troubleshooting	Chapter 18 pp. 159-164; Chapter 19 pp. 165-175	Weekly mastery	[]
W14 Dec 07 - Dec 12	C Programming I + Computer Architecture	Chapter 16 pp. 132-148; Chapter 17 pp. 149-156	Weekly mastery	[]
W15 Dec 14 - Dec 19	C Programming II + Embedded C + State	Chapter 17 pp. 149-156	Weekly mastery	[]



Week / Dates	Topic	Book	Due / Gate	Status
	Machines			

Winter Embedded Intensive

Week / Dates	Topic	Book	Due / Gate	Status
W16 Dec 21 - Dec 26	HOLIDAY LIGHT: STM32 Setup + GPIO + SWD Debug	Chapter 17 pp. 149-156	Weekly mastery	[]
W17 Dec 28 - Jan 02	HOLIDAY LIGHT: UART + Python Serial + Computer Applications	Chapter 16 pp. 132-148; Chapter 17 pp. 149-156	Weekly mastery	[]
W18 Jan 04 - Jan 09	I2C/SPI + Logic Analyzer + Sensors/Actuators	Chapter 16 pp. 132-148; Chapter 17 pp. 149-156	Embedded Interface Gate	[]

Term II - Applied Transition

Week / Dates	Topic	Book	Due / Gate	Status
W19 Jan 11 - Jan 16	CETa-Specific Compression: Cabling + Audio/Video + Optical	Chapter 9 pp. 70-78; Chapter 14 pp. 113-119; Chapter 21 pp. 182-187	Weekly mastery	[]
W20 Jan 18 - Jan 23	CETa-Specific Compression: RF + Telecom + Technician Procedures	Chapter 13 pp. 104-112; Chapter 14 pp. 113-119; Chapter 15 pp. 120-130; Chapter 22 pp. 188-193	Weekly mastery	[]
W21 Jan 25 - Jan 30	CETa Final Review I: Formula + Component + Diagram + Safety Gates	Targeted review / no major new chapter	CETa Review	[]
W22 Feb 01 - Feb 06	CETa Final Review II + Project 1 Design	Targeted review / no major new chapter	CETa Practice + Project Design	[]
W23 Feb 08 - Feb 13	CETa Exam Target + Project 1 Setup	Targeted review / no major new chapter	CETa Readiness + Project Setup	[]
W24 Feb 15 - Feb 20	Project 1 Hardware Bring-Up	Chapter 11 pp. 85-90; Chapter 18 pp. 159-164; Chapter 19 pp. 165-175	Weekly mastery	[]
W25 Feb 22 - Feb 27	Project 1 Sensor / Interface Integration	Targeted review / no major new chapter	Weekly mastery	[]
W26 Mar 01 - Mar 06	Project 1 Firmware Architecture	Chapter 17 pp. 149-156	Weekly mastery	[]
W27 Mar 08 - Mar 13	Python Hardware Test + Automated Logger	Chapter 22 pp. 188-193	Weekly mastery	[]
W28 Mar 15 - Mar 20	Fault Injection + Troubleshooting + Test Report	Chapter 18 pp. 159-164; Chapter 22 pp. 188-193	Root-Cause Report	[]
W29 Mar 22 - Mar 27	Instrumentation + Solder/Rework/Harness Role Proof	Chapter 14 pp. 113-119; Chapter 19 pp. 165-175; Chapter 20 pp. 176-180	Role-Proof Dossier	[]
W30 Mar 29 - Apr 03	Portfolio + Resume + Interview Green-Light	Chapter 21 pp. 182-187; Chapter 22 pp. 188-193	Career Final / Green-Light	[]
W31 Apr 05 - Apr 10	APPLICATION SPRINT + Interview Feedback Loop	Chapter 21 pp. 182-187	Application Sprint	[]



15. Rubrics

Weekly Lab / Practical - 100 points

Criterion	Points
Preparation & safety	15
Correct setup / method	20
Measurements / evidence	25
Troubleshooting / reasoning	25
Documentation / cleanup	15

Project 1 Capstone - 100 points

Criterion	Points
Requirements & architecture	15
Hardware integration / bring-up	20
Firmware structure & reliability	20
Python automation / data capture	15
Instrumentation & test evidence	15
Troubleshooting / root-cause stories	10
Documentation / demo / professionalism	5

Career Final - 100 points

Criterion	Points
Targeted resume	20
GitHub / portfolio proof	20
90-second transition story	15
Five technical/behavioral stories	20
Interview technical defense	15
Final green-light judgment / job filters	10

16. Student Contract

I understand that this program is self-directed and non-accredited. I will use the deadlines to create structure, but I will not treat a calendar date as proof of competence. I will not mark skills complete without evidence. I will prioritize safe low-voltage lab practice, honest self-assessment, college responsibilities and sustainable progress. When I reach the Week 30 green-light gate, I will begin applying rather than moving the finish line with optional skills.



Student: _____ Date: _____

Course start signature: _____ Completion date: _____